



AI Toolkit for Mech Dept.

AI

for Mechanical Engineering.

From CAD to Smart Manufacturing.

HOW MECHANICAL ENGINEERS CAN USE AI FOR

01 Design

02 CAD

03 Simulation

04 Manufacturing

05 Maintenance

06 Research

07 Data Analysis

08 Programming

09 Documentation

10 Productivity

BOTTOM LINE

AI is not replacing Mechanical Engineers.

Mechanical Engineers who use AI will increasingly outperform those who don't.

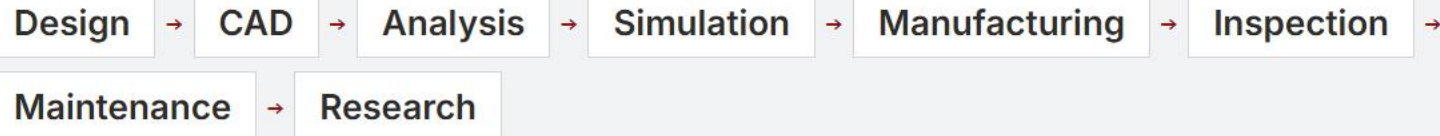
02 · ASK BEFORE YOU TELL

What does AI have to do with Mechanical Engineering?

ASK STUDENTS — HANDS UP

“Where do you think AI can be used in Mechanical Engineering?”

THEN SHOW THESE AREAS



THEN ASK

“Can AI help in all these stages?”

ANSWER

YES

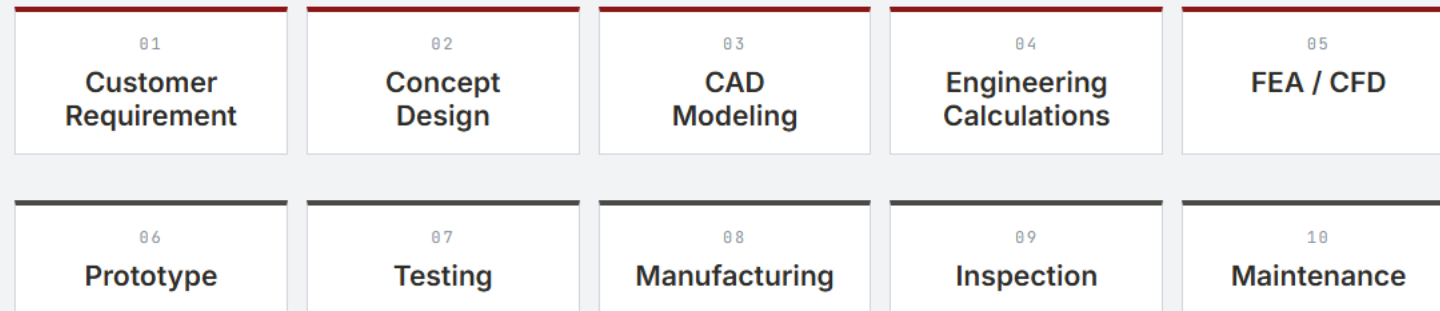
— but in different ways.

Each stage uses a *different* kind of AI.
Don't let the class assume one tool fits all eight.

Open with the question, not the definition — the class answers first, the map comes on slide 05.

03 • TRADITIONAL PRODUCT DEVELOPMENT

The workflow you already know.



↓ SEQUENTIAL — EACH STAGE WAITS FOR THE ONE BEFORE IT

HOLD THIS THOUGHT

None of these ten stages disappears. Every one of them contains repeated, searchable, documentable work — and *that* is where AI attaches.

Slide 04 shows where it attaches.

THE PROBLEM

Every stage requires:

- Time
- Engineering knowledge
- Manual work
- Repeated calculations
- Data analysis
- Documentation

Baseline first, tool second — the class must be able to name the stage before it names the tool.

04 · AI BECOMES THE ASSISTANT ACROSS THE WORKFLOW

Think of AI as your engineering copilot.

A MECHANICAL ENGINEER NORMALLY HAS TO

Search → Calculate → Design → Simulate → Analyse → Document

WITH AI

Ask → Generate → Analyse → Verify → Improve

KEY MESSAGE

AI doesn't replace the engineering workflow. It adds an intelligent assistance layer to it.

The verification rule that makes this safe is on slide 27. Don't skip it.

AI CAN HELP YOU

- Generate ideas
- Explain engineering concepts
- Write calculations / code
- Analyse data
- Automate repetitive tasks
- Generate design alternatives
- Summarize technical documents
- Assist with simulations
- Optimize designs

BUT

Engineer = Decision Maker
AI = Assistant

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05 · THE 8 AI AREAS EVERY MECHANICAL STUDENT SHOULD KNOW

Mechanical Engineering AI Map.

#	AREA	WHAT AI CAN HELP WITH
01	Learning	Understand concepts
02	Design	Generate design ideas
03	CAD	Assist modelling
04	Simulation	FEA / CFD assistance
05	Manufacturing	Optimization & automation
06	Maintenance	Predict failures
07	Data	Analyse engineering data
08	Research	Papers, reports & experimentation

WHY THIS SLIDE MATTERS

Students can use this as their AI roadmap.

Eight areas, one per column of your semester. Pick the row that matches this week's assignment — start there, not with the tool list.

SLIDE MAP

Learning & concepts → 08-10

Design & CAD → 11-14

Simulation → 15-16

Data & maintenance → 17-19

Manufacturing → 20-23

06 · AI TOOL CATEGORIES

Not all AI tools are the same.

1**General AI Assistants**

EXAMPLES

ChatGPT
Gemini
Claude
Microsoft Copilot

2**Engineering AI**

EXAMPLES

Autodesk Fusion
Generative Design
Ansys AI / SimAI
Altair
Siemens engineering AI

3**AI + CAD**

EXAMPLES

Fusion
SOLIDWORKS ecosystem
Creo
Siemens NX
AI CAD assistants

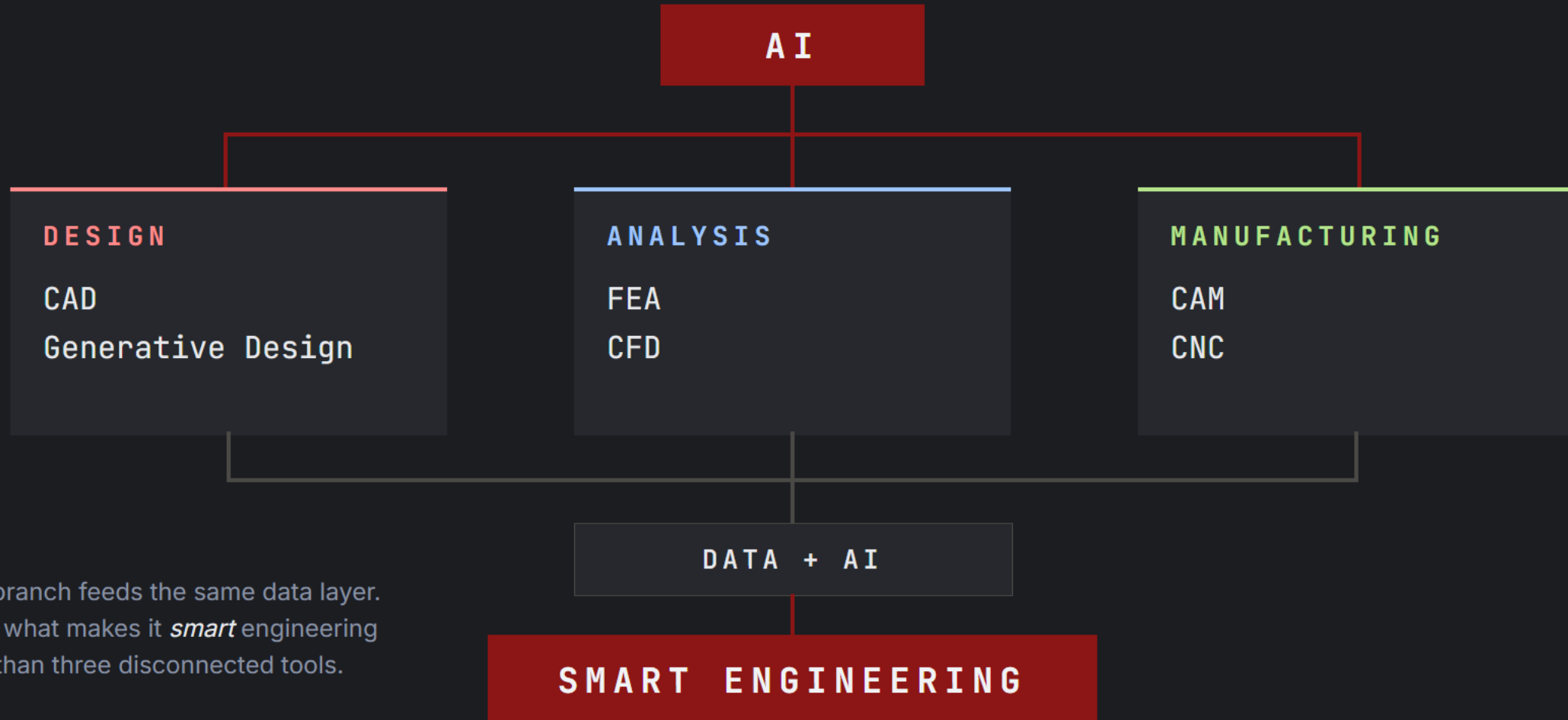
4**AI + Programming /
Data**

EXAMPLES

ChatGPT
Gemini
GitHub Copilot
MATLAB Copilot
Python

Choosing the wrong category is the most common student mistake — a chatbot is not a solver, and a solver will not explain a concept to you. Decision map: slide 29.

Three branches. One destination.



Every branch feeds the same data layer.
That is what makes it *smart* engineering
rather than three disconnected tools.

08 · YOUR FIRST AI USE CASE – OPEN A CHAT WINDOW

Type this prompt. 90 seconds.

prompt – ChatGPT / Gemini / Claude

```
# Copy this into any general AI assistant.  
"Explain the difference between stress and strain  
to a second-year mechanical engineering student  
using a bicycle frame example."
```

WHAT YOU SHOULD GET BACK

Simple explanation

Real-world example

Diagram concept

Formula

Practice questions

Quiz

DEBRIEF – 2 MINUTES

Compare answers with the person next to you. Did the AI use the same formula? The same units? Did either answer *invent* a number?

TOOLS FOR THIS TASK

TOOL	FREE	PAID
ChatGPT	Yes	Yes
Gemini	Yes	Yes
Claude	Limited	Yes
Microsoft Copilot	Plan dependent	Yes

A GOOD ENGINEERING PROMPT HAS

1. The **concept** to explain
2. The **audience level**
3. A **physical example** to anchor it
4. The **output format** you want

Drop any one of the four and the answer gets generic.

09 · THERMODYNAMICS & FLUID MECHANICS

Two subjects. Same prompt pattern.

THERMODYNAMICS

AI can help students:

- Understand thermodynamic cycles
- Explain entropy
- Compare Otto vs Diesel cycles
- Solve conceptual questions
- Generate practice problems
- Interpret P-V and T-S diagrams

→ Create revision notes

EXAMPLE PROMPT

"Explain the Diesel cycle step-by-step and compare it with the Otto cycle using a simple table."

FLUID MECHANICS

AI can help with:

- | | |
|----------------------|-------------------|
| → Bernoulli equation | → Pipe flow |
| → Reynolds number | → Boundary layers |
| → Flow regimes | → Pump concepts |
| → Turbines | → CFD preparation |

EXAMPLE PROMPT

"Explain Reynolds number using an automobile cooling system example."

IMPORTANT — AI SHOULD EXPLAIN AND ASSIST

Students must still own the chain of reasoning.

Formula → Assumptions → Calculation
→ Units → Result

10 · AI CAN ASSIST WITH

Machine design, step by step.

Design calculations

Shaft design

Bearing selection

Gear calculations

Springs

Bolts

Factor of safety

Material
comparison

WHY THIS PROMPT SHAPE

Forcing assumptions to the top makes the AI's reasoning auditable. You can then check each assumption against your standard before you ever look at the result.

example prompt - shaft design

```
"Help me design a shaft for a given torque and speed.  
First list the assumptions and equations,  
then show the calculation steps."
```

```
# Note the order: assumptions BEFORE numbers.  
# If it answers with a number first, ask again.
```

Try it in class: give the same torque and speed to three different assistants. Compare the assumptions each one silently made — that gap is the lesson.

GOLDEN RULE

Don't blindly copy the answer.

VERIFY

Units

Assumptions

Formulas

Engineering standards

11 • AI FOR CAD

AI + CAD = faster design exploration.

The modelling skill stays yours. What changes is how many options you can afford to look at before you commit.

WORTH KNOWING

Autodesk Fusion's Generative Design can generate multiple design outcomes based on geometry, performance and manufacturing constraints.

We unpack exactly how on slide 12.

AI CAN ASSIST ENGINEERS WITH

- 01 Design concepts
- 02 Feature suggestions
- 03 Modelling assistance
- 04 Design automation
- 05 Parametric modelling
- 06 Design optimization
- 07 Generative design

12 · TRADITIONAL VS GENERATIVE

Who generates the options?

TRADITIONAL DESIGN – THE ENGINEER LOOPS

Idea



CAD



Test



Modify



Test again



Modify

GENERATIVE DESIGN – THE ENGINEER SPECIFIES

ENGINEER PROVIDES

Loads

Material

Manufacturing Method

Constraints

Objectives



AI



Multiple Design Options



Engineer Selects



Validate

KEY MESSAGE

AI generates options. The engineer chooses and validates.



13 · EXAMPLE – LIGHTWEIGHT BRACKET

Five constraints. Many geometries.

REQUIREMENT – DESIGN A BRACKET THAT

- 01 Supports 500 N
- 02 Uses Aluminium
- 03 Must be lightweight
- 04 Must be manufacturable
- 05 Must satisfy safety constraints

AI / generative design can explore multiple geometries against exactly these five inputs — then hand you a set, not an answer.

STUDENT QUESTION – TURN TO YOUR NEIGHBOUR, 3 MINUTES

“Which design would you select?”

THEN DISCUSS – THE SELECTION CRITERIA

Weight

Strength

Manufacturing

Cost

Safety

Material

There is no single correct answer here – the point is that the engineer, not the software, owns the trade-off.

14 · TOOLS STUDENTS SHOULD EXPLORE

Two platforms, not six.

TOOL	MAIN USE	STUDENT ACCESS
Autodesk Fusion	CAD + Generative Design	Education / free options
SOLIDWORKS	CAD + design ecosystem	Mostly licensed
Siemens NX	Advanced CAD / PLM	Mostly licensed
PTC Creo	CAD + simulation	Mostly licensed
nTopology	Advanced optimization / lattice	Mostly paid
AI CAD copilots	Prompt / automation assistance	Varies

TEACHING RECOMMENDATION

Start students with Fusion + their existing CAD platform.

Rather than trying to teach six CAD systems in one semester.

READING THE ACCESS COLUMN

Licensing terms move faster than any slide deck. Check the current education offer before you promise a class access to anything on this list.

Full free-vs-paid picture across all tool categories: slide 28.

15 · AI + FINITE ELEMENT ANALYSIS / COMPUTATIONAL FLUID DYNAMICS

The solver stays. The loop gets shorter.

TRADITIONAL FEA

Geometry



Material



Mesh



Boundary Conditions



Solve



Results

AI CAN ASSIST WITH

- Simulation setup
- Meshing
- Result interpretation
- Design optimization
- Surrogate models
- Faster prediction

CFD APPLICATIONS

- Aerodynamics
- Heat transfer
- HVAC
- Turbomachinery
- Automotive
- Aerospace
- Flow optimization

THE CFD LOOP — BEFORE AND AFTER

Traditional: Run CFD → Wait → Analyse → Change geometry → Run again

AI-assisted: Existing simulation data → AI model → Rapid prediction → Engineer validates

READ THIS TWICE

AI simulation tools increasingly use trained models to predict physics results from previous simulation data — but they do **not** universally replace validated FEA/CFD or physical testing.

16 · TOOLS TO KNOW

Two very different shelves.

BEGINNER / STUDENT

Where you start.

ChatGPT

Gemini

MATLAB

Python

ENGINEERING SIMULATION

Where the physics lives.

Ansys

Ansys SimAI

SimScale

Altair HyperWorks

Siemens Simcenter

Neural Concept

WHAT THESE PLATFORMS TARGET

Some current engineering AI platforms specifically target:

→ Physics prediction

→ Simulation acceleration

→ Design optimization

That is a narrower promise than "AI does simulation". Read the promise carefully.

IMPORTANT DISTINCTION

AI chatbot ≠ FEA solver

A chatbot can help you *set up* and *interpret* a simulation. It does not solve the field equations. Never present a chatbot's numbers as simulation results.



17 · MECHANICAL ENGINEERS WRITE CODE — WITH HELP

MATLAB for computing. Python for everything after.

MATLAB IS USEFUL FOR

- Numerical analysis
- Control systems
- Signal processing
- Modelling
- Data analysis
- Optimization
- Engineering simulations

MATLAB Copilot provides generative-AI assistance for writing, explaining and debugging MATLAB code, as well as generating test cases.

EXAMPLE PROMPT

"Create MATLAB code to plot temperature variation with time and calculate the rate of change."

Pick one language this semester. The AI assistant removes the syntax barrier — it does not remove the need to read your own output critically.

WHY LEARN PYTHON?

- Engineering calculations
- Data analysis
- Automation
- Machine learning
- Predictive maintenance
- Optimization
- Visualization
- Sensor data

AI assistants can help students write *and* debug the code — the debugging half is where the learning happens.

```
# the mechanical engineer's Python stack
Python
├─ NumPy          # arrays, linear algebra
├─ Pandas         # tabular sensor data
├─ Matplotlib     # plots for reports
├─ SciPy          # integration, optimization
├─ Scikit-learn   # classical ML
└─ PyTorch        # deep learning
```



18 · IMAGINE YOU HAVE 10,000 SENSOR RECORDS

You are not reading those by hand.

machine_log.csv - 10,000 rows

Time	Temperature	Pressure	RPM	Vibration	Flow Rate
------	-------------	----------	-----	-----------	-----------

... 10,000 rows of this. Six columns. One question:
"What is this machine about to do?"

INSTEAD OF ANALYSING MANUALLY - AI CAN HELP

Detect patterns

Identify anomalies

Plot graphs

Find correlations

Predict trends

Build ML models

CAUTION

A correlation the AI finds in sensor data is a hypothesis, not a mechanism. You still have to explain it physically before you act on it.

TOOLS

Python

Excel + Copilot

ChatGPT

Gemini

MATLAB

Power BI + AI features

19 • PREDICTIVE MAINTENANCE

From “fix after failure” → “predict before failure”.

TRADITIONAL

Machine

**Failure**

Repair

WHY IT MATTERS

Unplanned downtime is the expensive failure mode. Prediction moves the repair into a scheduled window.

PREDICTIVE MAINTENANCE

Machine Sensors



Data

**AI****Failure Probability****Maintenance**

EXAMPLE

A motor produces unusual vibration.

AI identifies:

“Vibration pattern is different from normal operating conditions.”

Engineer investigates.

READ THE OUTPUT CAREFULLY

The AI said “different”. It did not say “broken”.

Anomaly detection flags deviation from a learned baseline. Diagnosing the cause — bearing, misalignment, imbalance, looseness — is still engineering work.

20 · SMART MANUFACTURING

Two extra links in the decision chain.

AI CAN SUPPORT

Production planning	Quality inspection	CNC optimization	Tool wear prediction
Defect detection	Process optimization	Predictive maintenance	Robotics

TRADITIONAL FACTORY

Machine → Human → Decision

SMART FACTORY

Machine → Sensors → Data → AI → Engineer → Decision

NOTICE WHAT DIDN'T CHANGE

The engineer is still the last box before the decision.

Sensors and AI were inserted *upstream* of the human, not in place of the human. That single structural fact is the difference between automation that works on a factory floor and automation that gets switched off in week three.

Ask the class: which of the eight tasks above could run with *no* engineer in the loop? Defend your answer.

21 · AI FOR QUALITY INSPECTION

Imagine a production line.

CAMERA CAPTURES

1000 components / hour

AI CHECKS FOR

Cracks

Surface defects

Wrong dimensions

Missing components

Scratches

Assembly errors

THE TWO ERRORS THAT MATTER

A vision model has two failure modes and they are not equally expensive: a **false accept** ships a cracked part to a customer; a **false reject** scraps a good one. Tune the threshold to the consequence, not to the accuracy score.

TOOLS / TECHNOLOGIES

Computer Vision

OpenCV

YOLO

Roboflow

Python

Industrial vision systems

22 · AI FOR CNC & CAM

Six parameters worth optimizing.

AI CAN HELP OPTIMIZE

- 01 Tool paths
- 02 Cutting parameters
- 03 Material removal
- 04 Tool life
- 05 Cycle time
- 06 Surface quality

STUDENT PROJECT IDEA

“Use historical CNC data to predict tool wear.”

THIS COMBINES

Mechanical Engineering

+

Python

+

Machine Learning

HOW TO SCOPE IT

One machine. One tool type. One material. Log spindle load and cycle count, label the point where the tool was changed, and predict that label. A project that fits one semester beats a project that fits a press release.



23 • CHOOSING THE RIGHT MATERIAL

The handbook becomes a shortlist engine.

TRADITIONAL

Requirement



Search Handbook



Compare Materials



Calculate



Select

AI-ASSISTED

Requirements



AI-assisted Search



Candidate Materials



Engineering Validation



Final Selection

COMPARE ON

Strength

Density

Cost

Temperature resistance

Corrosion

Manufacturability

24 · RESEARCH BECOMES FASTER

Eight things students can do today.

Understand research papers**Analyse datasets****Summarize papers****Explain equations****Compare papers****Create literature-review tables****Generate research questions****Improve technical writing****WORTH KNOWING**

Google has evolved NotebookLM into Gemini Notebook, with notebook-based source organization and study capabilities.

NON-NEGOTIABLE FOR RESEARCH WORK

A summary is not a citation. Open the paper, find the sentence, check the figure — then cite the paper, never the chatbot.

USEFUL TOOLS**Gemini Notebook**

source-grounded study notebooks

ChatGPT

explanation and drafting

Perplexity

sourced search

Elicit

literature review workflow

Semantic Scholar

paper discovery

25 · MECHANICAL ENGINEERS WRITE A LOT

Raw observations in. Structured report out.

AI CAN ASSIST WITH

Lab reports	Design reports	Project documentation
SOPs	Maintenance reports	Inspection reports
Meeting notes	Technical presentations	Emails

EXAMPLE

Give AI: **raw engineering observations** → AI produces:
structured technical report

BUT — THE ONE RULE FOR ALL WRITTEN OUTPUT

Never allow AI to invent measurements or test results.

DRAWINGS & DOCUMENTATION

From drawing → understanding.

AI can *potentially* assist with:

- Drawing interpretation
- Part descriptions
- BOM assistance
- Design documentation
- Requirement extraction
- Design review

FUTURE WORKFLOW

CAD + Drawing



AI



Design Information



BOM / Documentation / Review

26 · CHATGPT / GEMINI AS CALCULATION ASSISTANTS

Six steps. Ask for all six, in order.

STEP 01

Identify the formula

STEP 02

List assumptions

STEP 03

Substitute values

STEP 04

Calculate

STEP 05

Check units

STEP 06

Explain the answer

● ● ● example prompt – bending stress

"Calculate the bending stress for a shaft.
Show formula, units, assumptions and final result."

Naming the four outputs is what stops a one-line answer.

Class exercise: ask for the same bending stress twice, once in mm and once in m. If the two answers do not agree after conversion, you have just found the failure mode live.

BUT REMEMBER

**AI-generated
calculation $\neq$
engineering approval.**

WHERE THESE ASSISTANTS FAIL QUIETLY

- Silent unit conversion (mm vs m)
- An assumption you never stated
- A formula valid for a different geometry
- A rounded constant carried through
- Confident arithmetic that is simply wrong

All five look identical to a correct answer.
That is the whole problem – and slide 27 is the fix.

Never trust AI blindly.

AI → VERIFY → ENGINEER

CHECK — EVERY TIME, NO EXCEPTIONS

✓ Formula

✓ Units

✓ Assumptions

✓ Boundary conditions

✓ Material properties

✓ Standards

✓ Simulation settings

✓ Experimental data

GOLDEN RULE

**If the result affects
safety, AI must not
be the final
authority.**

WHAT “FINAL AUTHORITY” MEANS IN PRACTICE

A named engineer signs the calculation. Their name — not a model version — appears on the drawing, the report and the approval. AI output is an input to that signature, never a substitute for it.

28 • WHAT CAN STUDENTS START WITH?

Twelve tools, honestly labelled.

TOOL	FREE / FREE TIER	PAID	BEST FOR
ChatGPT	Yes	Yes	General engineering AI
Gemini	Yes	Yes	Study + research
Claude	Yes / limited	Yes	Writing / reasoning
Microsoft Copilot	Plan dependent	Yes	Productivity
Gemini Notebook	Availability dependent	Yes	Research / study
Python	Yes	—	Engineering programming
MATLAB	Limited	Yes	Engineering computing
Fusion	Education options	Yes	CAD / design
Ansys	Academic access	Yes	FEA / CFD
SimScale	Yes / limited	Yes	Cloud simulation
GitHub Copilot	Eligibility-dependent	Yes	Coding
nTopology	Academic options	Yes	Advanced optimization

IMPORTANT

Free vs paid changes frequently.

For student training, always check the current education / student licensing before promising access.

EXAMPLE

Autodesk documents educational access for Fusion and its generative-design workflow — that is the kind of page to verify before class, every term.

Green = generally accessible • Amber = conditional • Red = expect a licence.

29 · WHICH AI TOOL SHOULD I USE?

Find your task. Start there.

YOUR TASK	START WITH
Learn a concept	ChatGPT / Gemini
Solve / explain engineering problem	ChatGPT / Gemini
Research papers	Gemini Notebook / Perplexity / Elicit
CAD	Fusion / SOLIDWORKS / Creo / NX
Generative Design	Fusion
FEA	Ansys
CFD	Ansys / SimScale
MATLAB	MATLAB + Copilot

TAKE-HOME EXERCISE

Pick the row that matches this week's assignment.

Use that tool for one task only. Write down what it got right, what it got wrong, and which of the eight verification checks caught the error. Bring that note to the next session.

YOUR TASK	START WITH
Engineering programming	Python + AI
Data analysis	Python / Excel + AI
Predictive maintenance	Python + ML
Quality inspection	Computer Vision + Python
Reports	ChatGPT / Gemini
Presentations	ChatGPT / Gemini + presentation tools
Coding	GitHub Copilot / ChatGPT

HOW TO READ THIS MAP

Every row starts with a **task**, not a tool. That is deliberate. Students who start from “which AI is best?” pick a chatbot for a solver problem; students who start from the task land in the right category on the first try.

Whatever the row says – the output still goes through slide 27.

AI is not replacing Mechanical Engineers.

WHAT CHANGES

Mechanical Engineers who use AI will increasingly outperform those who don't.

THE FIVE THINGS TO WALK OUT WITH

01

AI attaches to every stage of the workflow

Design through maintenance — differently at each one.

02

Pick the category, then the tool

A chatbot is not a solver. A solver will not teach you.

03

AI generates options; you choose

Generative design hands you a set, not an answer.

04

Learn Python or MATLAB this semester

Data, maintenance and inspection all run through code.

05

Verify before you sign

Formula, units, assumptions, standards — every time.

THE RULE THAT OUTLIVES EVERY TOOL ON THESE SLIDES

AI → VERIFY → ENGINEER

If the result affects safety, AI must not be the final authority.



AI Tools in Mechanical Engineering

Explore, Learn, Build, Innovate



 Mechanical Area	 Tools to Introduce	 Free/Low-Cost Starting Point	 Professional/Advanced
 Learning	ChatGPT, Gemini		
 Research	Gemini Notebook, Perplexity, Elicit		
 Documentation	ChatGPT, Gemini		
 CAD	Fusion, SOLIDWORKS, Creo, NX		
 Generative Design	Fusion	 Education	
 FEA	Ansys, SimScale		
 CFD	Ansys, SimScale		
 Data Analysis	Python, Excel		
 Engineering Computing	MATLAB		
 Coding	Python + Copilot/ChatGPT		
 Predictive Maintenance	Python + ML		
 Quality Inspection	OpenCV, YOLO, Roboflow		
 Manufacturing	Fusion/CAM, Siemens, Ansys/Altair ecosystem		
 Research	Gemini Notebook, ChatGPT, Elicit		



= Good to go / Available



= Limited / Paid / Education option



= Not available / Not recommended



= Partial / Some features free